杭州电子科技大学学生考试卷（ A ）卷

考试课程	通信电路与系统	考试日期	2023 年 6 月 日	成 绩			
课程号	A040095s	教师号		任课教师姓名	林弥、陈瑾、李航、刘艳、高一强、张伟、周涛、苏江涛、刘国华、关阳阳、孙朋飞		
姓名		学号		年级		专业	

题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题	第七大题
得分							

I. Single Choice (3' each)

1 Which component should not be contained in modern communication system?(C)

A. Modulator B. Power Amplifier

C. Microphone D. Oscillator

2. Which of the following statements about Class C PA is correct (D)

A. Class C PA has conduction angle higher than 90°.

B. Only BJT transistors can operate in Class C working status.

C. The role of Class C PA in communication system is only amplifying signal.

D. None of above

3. For LC series resonant circuit with resonance frequency f_0, when operate frequency is at f_0, which of the following statement is correct? (C)

A. The circuit shows Capacitive

B. The circuit shows Inductive

C. The circuit has minimum impedance

D. The circuit voltage reaches the maximum

4. A power amplifier is biased with 60° conduction angle, if the input signal is a sinusoidal signal, the collector current would be: (C)

A. sinusoidal signal B. Square wave signal

C. Pulse cosine signal D. Depending on the DC Bias

5. What is the following statement about Quartz crystal is correct (C)

A. It has high temperature coefficient

B. When used in oscillator, it can be seen as capacitor at working status

C. It has *high Q-factor*.

D. It has high order even harmonic resonant frequencies.

6. What is the following statement about oscillator is correct (B)

A. Negative feedback satisfy the oscillation startup condition

B. Oscillator is used both in transmitters and receivers.

C. Hartley oscillator has higher oscillation frequency.

D. Colpitts Oscillator is easier to start oscillating.

7 When a typical DSB signal pass a mixer, what will happen? (B)

A. The envelope shape will change

B. The frequency will change

C. The bandwidth will change

D. None of above

8. A broadcasting radio signal at 1090 kHz is received by a receiver with another signal at 1323 kHz is at the same time. Suppose intermediate frequency is 465KHz, Why does the 1323 kHz signal appear? (C)

A. Mirror image interference B. IF interference

C. Higher order side-channel interference D. Random Noise

9. For standard FM wave, which of the following statement is correct? (B)

A. The carrier amplitude will change with the modulated signal

B. The carrier frequency will change with the modulated signal

C. The carrier frequency and amplitude both change with the modulated signal

D. The carrier frequency and amplitude both remain constant

10. Which of the following component is not a part of super heterodyne receiving system? (D)

A. Local Oscillator B. Mixer

C. Intermediate frequency amplifier D. Modulator

II. Calculation

1. (13') Suppose you have designed a resonant power amplifier, but found that the output power cannot meet the design specifics. The DC bias is fixed, and decrease or increase the input signal's amplitude don't help. After further investigation you found that I_{c0} is higher than expected.

(1) What could be reasons of low output power? How to solve it?

(2) You have finally managed to get the maximum output power. At this working status, suppose $Ec = 28V$, current conduction angle $\theta = 70°$, $\alpha 0\ (70°) = 0.253$, $\alpha 1\ (70°) = 0.436$, collector current pulse maximum value $Ic_{max} = 1.2A$, collector voltage utilization factor $\xi=0.8$. Please calculate output power Pout, power of power supply P_{DC}, collector efficiency η_c and collector equivalent load resistor Rc.

(3) When this PA is used in the wireless communication system, the output efficiency and power is smaller than expected. Please talk about the influences on environment and sustainable development.

(4) How should we solve the above problem? (Including adjust DC bias)

参考答案：

1 (2') 输出功率不够， I_{c0} 很大，说明放大器工作在欠压态，增加负载电阻可以使晶体管工作在临界态，输出功率增加。

2. (4')

$U_{c1m} = \xi \times E_c = 0.8 \times 28 = 22.4V$ $\quad$ $I_{c1m} = a_1(\theta) \times I_{cmax} = 0.436 \times 1.2 \approx 0.523A$

$P_o = \frac{1}{2} I_{c1m} U_{c1m} = \frac{1}{2} \times 22.4 \times 0.523 \approx 5.86W$ (1')

$I_{c0} = \alpha_0(\theta) \times I_{cmax} = 0.253 \times 1.2 \approx 0.304\ A$ $\quad$ $P_s = E_c \times I_{c0} = 28 \times 0.304 \approx 8.5\ W$ (1')

$\eta_c = \frac{P_o}{P_s} = \frac{5.86}{8.5} \approx 68.9\%$ (1')

$R_c = \frac{U_{c1m}}{I_{c1m}} = \frac{22.4}{0.523} \approx 42.8\Omega$ (1')

3. (5')功率放大器的作用原理是利用直流电源所供给的直流功率,使之转变为交流信号功率。如果效率过低，会导致部分功率以热能的形式消耗，甚至需要利用额外的散热系统，进一步降低了系统整体效率，导致消耗的能量过大，与我国碳达峰碳中和的整体发展目标不符。

4.(2') 可以通过调整晶体管工作状态来实现，比如 1) 选择合适的导通角,2)使功率放大器尽量靠近临界状态,4)选择合适的匹配网络,4) 可选择 F 类、E 类功率放大器。

（以上 4 点，答出两点以上或者其它合理方式即可给满分 2 分）

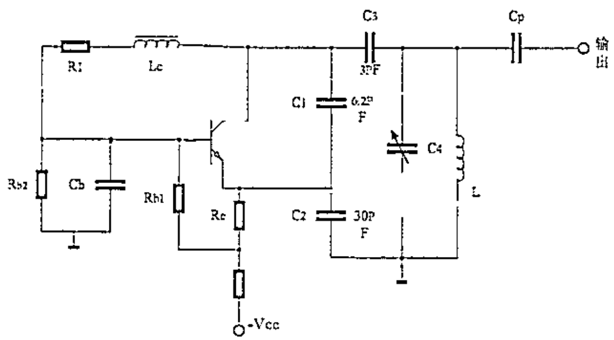
2. (12') The oscillation circuit is shown in below where Lc is the RF choke, L = 1.5uH, C_p is output couple capacitor. The oscillation frequency is 49.5MHz.

(1) Explain the purpose of R_{b1}, R_{b2}, R_e, C_1, C_2, C_3, C_4, C_b and L.

(2) Draw the AC equivalent circuit(Only contain the components contribute to resonant circuit).

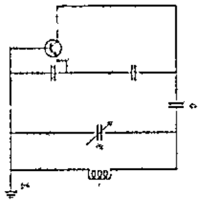
(3) Calculate the value of C_4(assume Inter-electrode capacitor is 0).

(4) If this circuit fails to oscillate, how should we adjust this circuit?



1）R_{b1}和 R_{b2}是基极偏置电阻，R_e是射极偏置电阻(1’)；C_1、C_2、C_3、C_4、L是振荡回路的元件(1’)，C_b是基极旁路电容(1’)。

（2）交流等效电路如图所示(3’)



（3）$f_0 \approx \dfrac{1}{2\pi\sqrt{L(C_3+C_4)}}$ (1’)

$$49.5\times10^6 \approx \frac{1}{2\pi\sqrt{1.5\times10^{-6}(C_3+C_4)}}$$

解得$(C_3+C_4)=6.91\times10^{-12}pF$ (2’)

$C_4=(6.91-3)pF=3.91pF$ (1’)

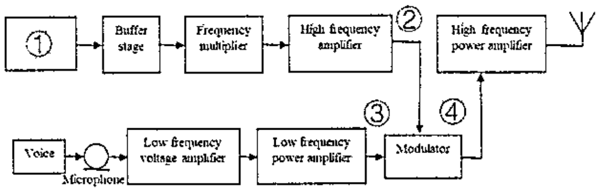
（4）若电路不起振，可以改变偏置或加大 C_4 或者调整 C1，C2,来调节反馈系数 (2’)

3. (10’)A standard AM modulation transmitter is shown below. The transmitted signal at position④ has the expression of $u=[2\cos(4\pi\times10^6t)+0.1\cos(3996\pi\times10^3t)+0.1\cos(4004\pi\times10^3t)]V$

1) Please fill in the component’s name at position①

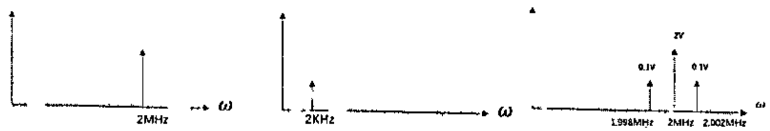
2) Suppose the modulator is lossless, please draw frequency domain spectrum at ②③④.

3) Please calculate signal u frequency bandwidth B, the modulation index m_a, the power of sideband frequency and the whole power with unit resistor.



参考答案： 1) Oscillator 或意思相近中英文表述（1'）

2)



上图分别为 2，3，4 处的频谱 (3')

注：如考生仅给了时域波形，在时域波形完全正确的情况下可酌情给 1 分

3) u 的数学表达式为:

$u(t)=2[1+0.1\cos(2\pi\times2\times10^3t)]\cos(2\pi\times2\times10^6t)V$

$m_a=0.1$ (2')

注：如数学表达式未写，m_a计算正确，可直接给 2 分

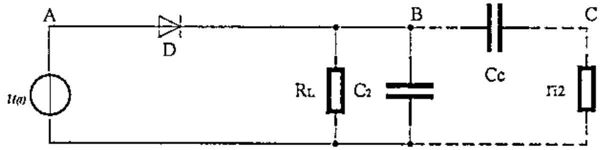
$P_c=\dfrac{1}{2}\dfrac{U_{cm}^2}{R_L}=2W$ (1)

$P_1=P_2=\dfrac{1}{2}(\dfrac{m_a}{2}U_{cm})^2\dfrac{1}{R_L}=\dfrac{1}{4}m_a^2P_c=0.005W$ (1')

$P=P_c+P_1+P_2=(1+\dfrac{m_a^2}{2})P=2.01W$ (1')

B = 2F = 4KHz (1)

4. (13') The circuit of large signal peak envelope detector is shown in Figure below, given the frequency of carrier wave is 465kHz, the highest frequency of modulation signal F=1000Hz, the diodes internal resistor $R_d = 100\Omega$, $R_L=5k\Omega$, $C_2=0.01uF$.



(1) If the detector is connected to the next stage circuit by a large capacitor C_c, $r_{i2} = 50k\Omega$, to avoid the distortion, what is the maximum modulation index m_a can be?

(2) If the input signal is $u(t) = 2\cos(2\pi \times 465 \times 10^3 t) + 0.3\cos(2\pi \times 467 \times 10^3 t) + 0.3\cos(2\pi \times 463 \times 10^3 t)V$ Suppose the detection efficiency is 1, draw the waveform at A,B and C.

参考答案:

(1) 为了不产生负峰切割失真, 当$R_L = 5k\Omega$时,

$m_{amax} \le \frac{R_\Omega}{R} = \frac{r_{i2}}{R_L r_{i2}} = \frac{50}{5+50} = 0.91$ （3'）

为了不产生惰性失真, 应满足不等式

$$R_L C\Omega_{max} \le \frac{\sqrt{1-{m_{amax}}^2}}{m_{amax}}$$

$R_L C$乘积为$5 \times 10^3 \times 0.01 \times 10^{-6} = 5 \times 10^{-5}$。 （3'）

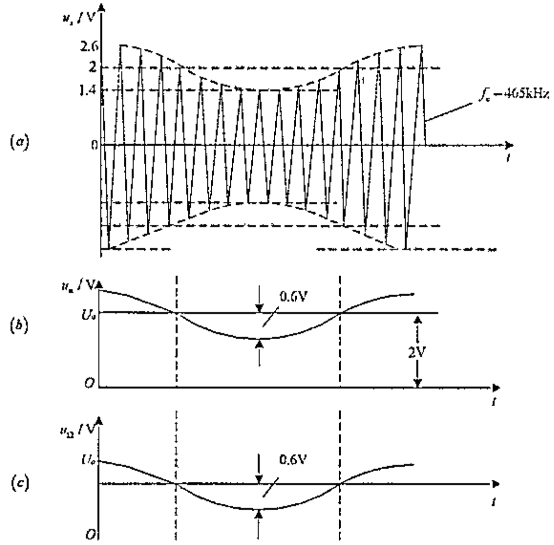
所以为了避免上述两种失真，允许的最大调制系数$m_{amax} = 0.91$ （1'）

(2) 该信号的调制系数 $m_a = 0.3$, F = 2KHz.

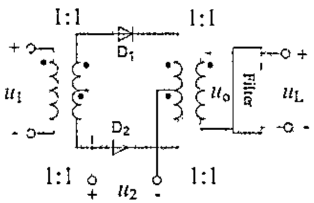
此时 $\frac{\sqrt{1-{m_{amax}}^2}}{m_{amax}\Omega_{max}} = 25.4\times 10^{-5}$ ，无惰性失真

m_a<0.91, 无负峰切割失真

波形如下图所示，每个波形各 2 分



5. (12') The diode multiplier circuit is shown below. Assume the I-V curve characteristics of all diodes are exactly the same and u_2 is much bigger than u_1. Please analyze the output signal u_o and u_L expression, the function of the circuit, and the type of filter under below input signals.



(1) $u_1 = u_{1m}\cos\Omega t\cos\omega_c t$，$u_2 = u_{2m}\cos\omega_c t$，$\omega_c \gg \Omega$

(2) $u_1 = u_{1m}(1+m_a\cos\Omega t)\cos\omega_c t$，$u_2 = u_{2m}\cos\omega_c t$，$\omega_c \gg \Omega$

参考答案：

(1) $u_1 = u_{1m}\cos\Omega t\cdot\cos\omega_c t$，$u_2 = u_{2m}\cos\omega_c t$ （大信号），$\omega_c \gg \Omega$时，

$u_o = u_{1m}\cos\Omega t\cdot\cos\omega_c t\cdot 2g_D\left(\frac{1}{2}+\frac{2}{\pi}\cos\omega_c t-\frac{2}{3\pi}\cos 3\omega_c t+\dots\right)$ (2')

其中，$\frac{2}{\pi}u_{1m}\cos\Omega t\cos^2\omega_c t=\frac{1}{\pi}u_{1m}\cos\Omega t+\frac{1}{\pi}u_{1m}\cos 2\omega_c t$，

经低通滤波器(1') 取出原调制信号$u_L=\frac{2}{\pi}u_{1m}\cos\Omega t$，(2') 实现双边带调幅波的同步解调(1')。

(2) $u_1 = u_{1m}(1+m_a\cos\Omega t)\cos\omega_c t$，$u_2 = u_{2m}\cos\omega_c t$ （大信号），$\omega_c \gg \Omega$时，有

$u_o = u_{1m}(1+m_a\cos\Omega t)\cos\omega_c t\, 2g_D\left(\frac{1}{2}+\frac{2}{\pi}\cos\omega_c t-\frac{2}{3\pi}\cos 3\omega_c t+\dots\right)$ (2') 其中

$\frac{2}{\pi}u_{1m}(1+m_a\cos\Omega t)\cos^2\omega_c t=\frac{1}{\pi}u_{1m}+\frac{1}{\pi}u_{1m}m_a\cos\Omega t+\frac{1}{\pi}u_{1m}\cos 2\omega_c t+$

$\frac{u_{1m}}{2\pi}\cos(2\omega_C+\Omega)t+\frac{u_{1m}}{2\pi}\cos(2\omega_C-\Omega)t$

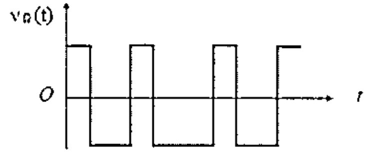
经低通滤波器(1') 取出原调制信号$u_L=\frac{2}{\pi}u_{1m}m_a\cos\Omega t$ (2')，实现普通调幅波的同步解调(1')。

注：上述u_o and u_L的表达式中，只要包含的频率分量正确即可给分。

6. (10') The modulation signal $V_\Omega(t)$ is known to be a rectangular wave as shown below. Try to:

(1)Write the expression the instantaneous frequency offset Δw and the instantaneous phase offset $\Delta\theta(t)$

(2) Draw the instantaneous frequency offset Δw and the instantaneous phase offset $\Delta\theta(t)$

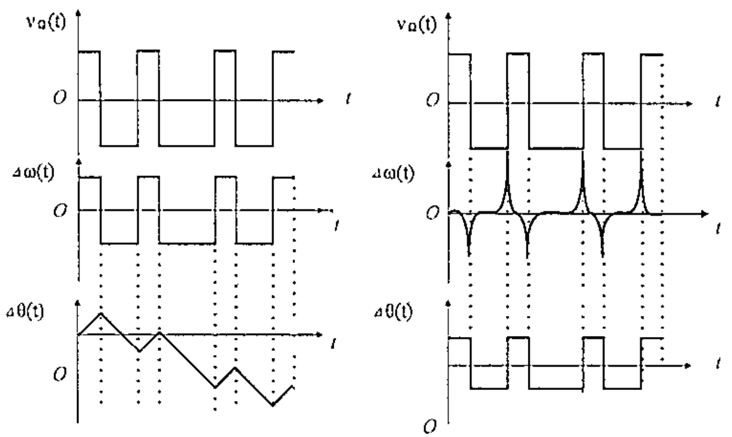


参考答案：

(1) FM时，瞬时频率的变化量$\Delta\omega\propto v_\Omega(t)$ (1')，瞬时相位的变化量$\Delta\theta\propto\int v_\Omega(t)\cdot dt$ (1')；

PM时，瞬时频率的变化量$\Delta\omega\propto\frac{dv_\Omega(t)}{dt}$ (1')，瞬时相位的变化量$\Delta\theta\propto v_\Omega(t)$ (1')

(2) FM PM



给分标准：

1. 单给出 FM 或者 PM 调制下 Δw 和 $\Delta\theta(t)$ 的，每个 2 分，共 4 分；
2. 同时给出 FM 和 PM 调制下 Δw 和 $\Delta\theta(t)$ 的，额外给 2 分；
3. 所给图形能定性反映出积分/微分关系即可酌情给分，不要求学生进行准确定量分析（例如 FM 相移曲线不要求形状和答案完全一致）

杭州电子科技大学学生考试卷（ B ）卷

考试课程	通信电路与系统	考试日期	2023 年 月 日	成 绩	
课程号	A040095s	教师号		任课教师姓名	林弥、陈瑾、李航、刘艳、高一强、张伟、周涛、苏江涛、刘国华、关阳阳、孙朋飞

题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题	第七大题
得分							

I. Judgment questions (3' each)

1. When the operation frequency is lower than the resonance frequency, a series LC resonant circuit shows capacitive.

( )

2. Television, AM/FM radio and mobile communications belong to millimeter wave (EHF) communications.

( )

3. Hartley circuit is easier to adjust frequency than Colpitts circuit.

( )

4. The frequency shift in an FM signal is related to the frequency of modulation signal.

( )

5. For Class C power amplifier, the conduction angle 2θ is less than $\pi/2$.

( )

6. Overtone crystal oscillator includes the fundamental frequency and its all-integer multiples.

( )

7. Compared with AM, FM signals are power and bandwidth saving.

( )

8. DSB transmitter saves more transmitting power than standard AM transmitter, because it has only half bandwidth of standard AM transmitter.

( )

9. Paralleling a resistor between two ends of LC resonant circuit will reduce the quality factor.

( )

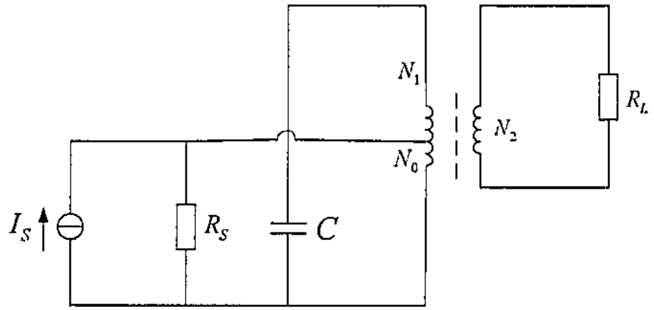
10. Angle modulation circuit is a linear spectrum shifting circuit.

( )

II. Calculation

1. (8') As shown in the figure, the resonant circuit f_0 = 600 kHz, Q_0 = 120, the access factor in primary coils $n_1 = N_1/N_0 = 3$, the access factor between primary and secondary coils $n_2 = N_1/N_2 = 9$, C = 180 pF, R_S = 20 kΩ and R_L = 1.5 kΩ.

(1) Calculate the inductor L and the resonant resistance R_0 (R_L and R_S are negligible).
(2) Calculate the loaded quality factor Q_L and bandwidth $B_{0.7}$.

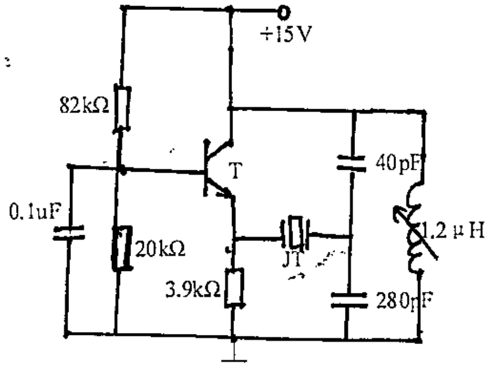


2. (8') A resonant power amplifier with operating at critical point state. E_C = 18 V, current conduction angle θ = 70°, α_0 (70°) = 0.253, α_1 (70°) = 0.436, U_{ces} = 3.3V and I_{cmax} = 4.6A.

(1) Please calculate the DC power P_{DC}, power dissipation at the collector P_c, collector efficiency η_c and optimal load resistor R_{cp}.
(2) To potentially prevent the transistor from burning out, what should we do for the equivalent load resistor?

3. (6') The oscillation circuit is shown below.

(1) Draw its AC equivalent circuit
(2) Is this crystal oscillator in parallel mode or series mode?
(3) Explain the roles of crystal.
(4) Calculate the feedback factor F of the circuit.



4. (15') The spectrum diagram of an AM signal is shown in Fig. (a)

(1) Please give the time-domain expression of this AM wave.
(2) Please calculate the whole power with unit resistor, and the bandwidth B.
(3) Please give the advantage and disadvantage of this kind of AM.
(4) Fig. (b) is the compositions diagram of AM receiving system. What do (a), (b) and (c) represent? If f_L=2465kHz, calculate the down-mixing f_I.

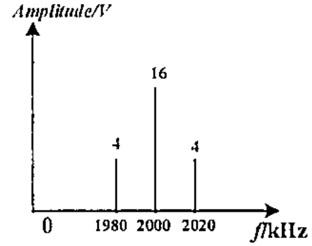


Fig. (a)

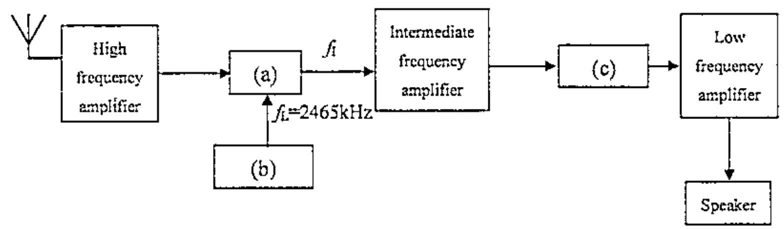


Fig. (b)

5. (16')

(1) Please draw the basic circuit of the diode large signal peak envelope detector.
(2) If $R_L = 8\ \mathrm{k\Omega}$, $m_a = 0.4$, $f_c = 600$ kHz and the highest frequency of modulation signal $F =$ 280 Hz, how is the capacitor chosen to prevent the diagonal distortion?
(3) If the detection efficiency $\eta_d = 0.97$, calculate the maximum equivalent input resistance R_{in}.
(4) How is the input resistance of next-stage low-frequency amplifier chosen to prevent negative peak clipping distortion?
(5) To demodulate the DSB and SSB signals, what kind of circuit can be selected? and draw the block diagram of its most commonly used circuit.

6. (14') The frequency of modulation signal is 300 Hz, its amplitude is 2.6 V and the angle modulation factor is 50.

(1) For FM signals, write the frequency modulation factor m_f and calculate the bandwidth;
(2) For PM signals, write the phase modulation factor m_p and calculate the bandwidth;
(3) When the frequency of modulation signal is reduced to 200 Hz and its amplitude is increased to 3.8 V, calculate the updated modulation factor and bandwidth of the FM and PM signals, respectively;
(4) Please draw the block diagram of slope frequency discriminator.

7. (3') Please list three interferences produced by the nonlinearity of mixer on communication environment.

B卷答案

I: 1,(√) 2,(X) 3,(√) 4,(X) 5,(X)

6,(X) 7,(X) 8,(X) 9,(√) 10,(X)

II: 1. $L=\frac{1}{(2\pi f_0)^2 C}=\frac{1}{(2\pi\times 600\times10^3)^2\times180\times10^{-12}}=391.3\,\mu H$ ①′

$R_0=Q_0/2\pi f_0 C=176.93\,k\Omega$ ①′

$R_\Sigma=R_0\,//\,R_S'\,//\,R_L'$ ①′

$R_S'=n_1^2R_S=9\times20\times10^3=180\,k\Omega$ ①′

$R_L'=n_2^2R_L=81\times1.5\times10^3=121.5\,k\Omega$ ①′

$\therefore R_\Sigma=51.45\,k\Omega$ ①′

$Q_L=\frac{R_\Sigma}{\omega_0 L}=29.4$ ①′

$B_{0.7}=\frac{f_0}{Q_L}=\frac{600\times10^3}{29.4}=20.41\,kHz$ ①′

2. 1) $P_{DC}=E_C I_{C0}=E_C\cdot I_{Cmax}\cdot\alpha_0(70^\circ)=18\times4.6\times0.253=20.95W$ ①′

$U_{cm}=E_C-U_{CES}=18-3.3=14.7V$ ①′

$P_0=\frac{1}{2}U_{cm}\cdot I_{am}=\frac{1}{2}U_{cm}\cdot I_{cmax}\cdot\alpha_1(70^\circ)=14.74W$ ①′

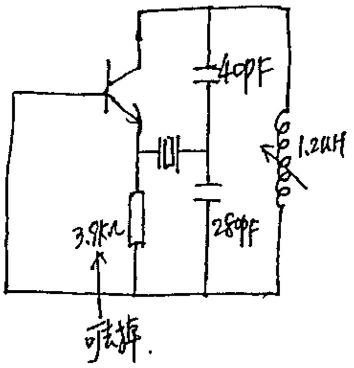
$\therefore P_C=P_{DC}-P_0=20.95-14.74=6.21W$ ①′

$\eta_C=\frac{P_0}{P_{DC}}=70.3\%$ ①′

$R_{cp}=\frac{U_{cm}^2}{2P_0}=7.3\Omega$ ①′

2) 我们应该避免等效阻抗太小。

3. (1)



③′

(2) 串联型晶体振荡器 ①′

(3) 晶体等效于短路元件 ①′

(4) $F = \frac{U_f}{U_o} = \frac{\frac{1}{j\omega_0 280pF}}{\frac{1}{j\omega_0 40pF} + \frac{1}{j\omega_0 280pF}} = 0.12$

4. 1) $u_{AM}(t) = 20(1 + 0.5\cos(2\pi \times 2\times10^4 t))\cos(2\pi \times 2\times10^6 t)$ (V)

①′ ①′ ①′ ①′

2) $P_c = \frac{1}{2}\frac{U_{cm}^2}{R} = \frac{1}{2}\times16^2 = 128\ W$ ①′

$P_{side} = \frac{1}{2}m_a^2 P_c = \frac{1}{2}\times0.25\times128 = 16W$ ①′

$P_{总} = P_c + P_{side} = 144W$ ①′

$B = 2\times2\times10^4 = 40kHz$ ①′

4) (a) 混频器 ①′ (b) 本地振荡器 ①′ (c) 检波器 ①′

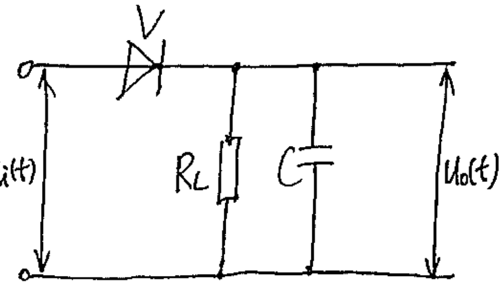
$f_I = |f_L - f_c| = 2465kHz - 2000kHz = 465kHz$ ②′

3) 优点：设备简单，易于接收 ①′

缺点：调制功率效率低 ①′

2)

5. (1)



③'

(2) $R_L C \gg \frac{1}{\omega_c} \Rightarrow C \gg \frac{1}{R_L \omega_c} = \frac{1}{8\times10^3\times 2\pi}$

$\doteq 33.2PF$

$C < \frac{\sqrt{1-m_a^2}}{m_a}\times\frac{1}{F\cdot R_L\cdot 2\pi}$

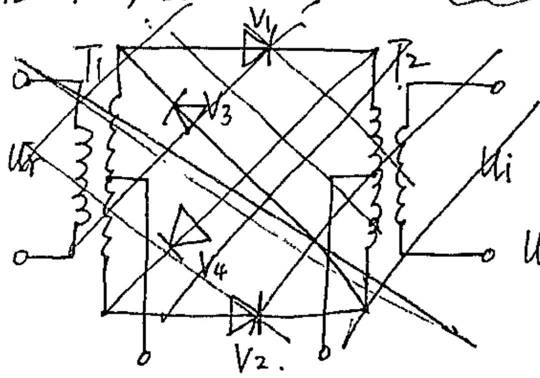
$= \frac{\sqrt{0.84}}{0.4\cdot 2\pi}\times\frac{1}{280\times8\times10^3} = 0.163uF$ ②

$\therefore 33.2PF \ll C < 0.163uF$

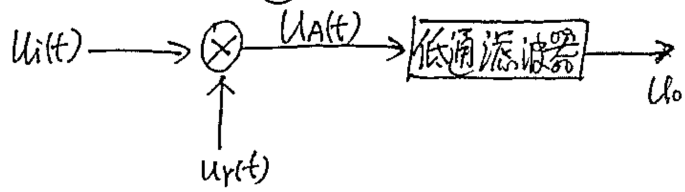
(3) $R_{in} \approx \frac{R_L}{2\eta_d} = \frac{8\times10^3}{2\times0.97} = 4123.7\Omega$ ②'

(4) $m_a \le \frac{R_i}{R_i+R_L} \Rightarrow R_i \ge \frac{m_a R_L}{1-m_a} = \frac{0.4\times8\times10^3}{1-0.4} = 5.33k\Omega$ ②'

(5) (二极管环形乘法器) ~~电路~~ 同步检波电路 ①'



④'



6. (2) $m_{f1} = \frac{K_f U_{\Omega1}}{\Omega_1}$, $m_{f2} = \frac{K_f U_{\Omega2}}{\Omega_2}$ $\therefore \frac{m_{f1}}{m_{f2}} = \frac{200\times2.6}{300\times3.8} = 0.46$ $\therefore m_{f2} = 108.7$ ②

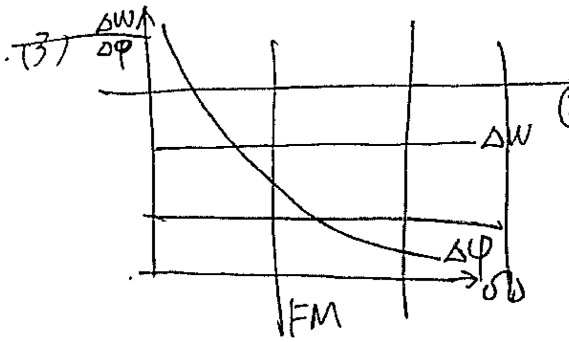
$B_{FM} = 2(m_{f2}+1)\times200 = 43.9KHz$ ①'

$m_{p1} = K_p U_{\Omega1}$, $m_{p2} = K_p U_{\Omega2}$ $\therefore \frac{m_{p1}}{m_{p2}} = \frac{2.6}{3.8}$ $\therefore m_{p2} = 73.53$ ②'

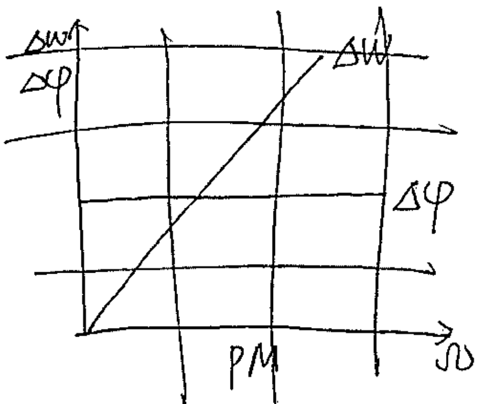
①'

$B_{PM} = 2(m_{p2}+1)F = 29.8kHz$

(3)



②'



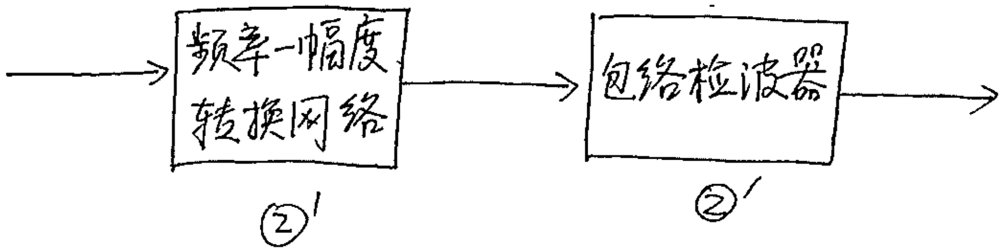
②'

解. (1) $m_{f1} = 50$ ①' $B_{FM} = 2(m_{f1}+1)\times300 = 30.6KHz$ ①'

(2) $m_{p1} = 50$ ①' $B_{PM} = 2(m_{p1}+1)\times300 = 30.6KHz$ ①'

3)

(4)



7. 1) 组合频率干扰（干扰哨声）

2) 副波道干扰　　　③′ 列出3个即可

3) 交叉调制干扰

4) 互调调制干扰

4)